

Lecture 12

Lur'e-Postnikov Lyapunov Functions

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The gradient method is relatively easy to analyze since it only requires quadratic Lyapunov functions and pointwise quadratic constraints. Specifically, we only need to construct a dissipation inequality in the form of $V(\xi_{k+1}) - \rho^2 V(\xi_k) \leq S(\xi_k, w_k)$ (or an expected version) with V being quadratic and S being smaller or equal to 0 for all k . Then we immediately have $V(\xi_{k+1}) \leq \rho^2 V(\xi_k)$. As a matter of fact, one can prove the linear convergence of the gradient method by only exploiting the one-point convexity of f . The convexity of f is not really needed. That is not the case for Nesterov's method.

Nesterov's method is more difficult to analyze. One reason is that more advanced Lyapunov functions and more sophisticated supply rates are required to exploit the properties of f . Just imagine that we still use the sector bound condition to analyze Nesterov's method. Hence we can set $X = \begin{bmatrix} C^\top & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} 2mLI & -(m+L)I \\ -(m+L)I & 2I \end{bmatrix} \begin{bmatrix} C & 0 \\ 0 & I \end{bmatrix}$ and have $S(\xi_k, w_k) \leq 0$. If we use this X to formulate the LMI, we will see that the testing condition is not feasible for the rate we want to test. Therefore, the sector bound condition is too conservative for Nesterov's method. Notice the key idea behind the dissipation inequality framework is to approximate $w_k = \nabla f(v_k)$ using some supply rate conditions. For Nesterov's method, we will need some supply rate conditions that can exploit the convexity better and give us Lyapunov functions in more general forms. In this lecture, we will talk about the Lur'e-Postnikov Lyapunov function approach for Nesterov's method.

12.1 Iteration Complexity of Nesterov's Method

The convergence rate ρ^2 naturally leads to an iteration number T guaranteeing the algorithm to achieve the so-called ε -optimality, i.e. $\|x_T - x^*\|^2 \leq \varepsilon$ or $f(x_T) - f(x^*) \leq \varepsilon$.

Based on the rate bound $c\rho^{2k}$, if we choose $T = \left\lceil \frac{\log(c/\varepsilon)}{-\log(\rho^2)} \right\rceil = O\left(\frac{\log(c/\varepsilon)}{1-\rho^2}\right)$, we guarantee the ε -optimal solution. The number T just gives the “ ε -optimal iteration complexity”. It is straightforward to verify that the convergence rate bound that we obtained for the gradient method can be converted to an iteration complexity $T = O\left(\frac{L}{m} \log\left(\frac{1}{\varepsilon}\right)\right)$.

Nesterov's method improves the iteration complexity from

$$O\left(\frac{L}{m} \log\left(\frac{1}{\varepsilon}\right)\right) \quad \text{to} \quad O\left(\sqrt{\frac{L}{m}} \log\left(\frac{1}{\varepsilon}\right)\right).$$

This improvement is significant. Just consider $\frac{L}{m} = 10000$. Then $\sqrt{\frac{L}{m}} = 100$. Hence the iteration-complexity scalings differ by a factor of 100 in this case. The convergence

rate corresponding to this iteration complexity is $\rho^2 = 1 - \sqrt{\frac{m}{L}}$. When f is L -smooth and m -strongly convex, Nesterov's method with $\alpha = \frac{1}{L}$ and $\beta = \frac{\sqrt{L}-\sqrt{m}}{\sqrt{L}+\sqrt{m}}$ satisfies a convergence bound $f(x_k) - f(x^*) \leq c(1 - \sqrt{\frac{m}{L}})^k$ where c is a constant. If we only use $X = \begin{bmatrix} C^\top & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} 2mLI & -(m+L)I \\ -(m+L)I & 2I \end{bmatrix} \begin{bmatrix} C & 0 \\ 0 & I \end{bmatrix}$ to form the supply rate function, the resultant LMI is not feasible with $\rho^2 = 1 - \sqrt{\frac{m}{L}}$. Now we will show how to analyze Nesterov's method by modifying the dissipation inequality and constructing the so-called Lur'e-Postnikov Lyapunov function.

12.2 Lur'e-Postnikov Lyapunov Functions

Although quadratic Lyapunov functions are not enough for proving the accelerated rate of Nesterov's method, we can use a Lyapunov function in the form of $(\xi_k - \xi^*)^\top P(\xi_k - \xi^*) + f(x_k) - f(x^*)$ to fix the issue. This type of Lyapunov function is exactly the so-called "Lur'e-Postnikov Lyapunov function" in the controls literature. The quadratic term $(\xi_k - \xi^*)^\top P(\xi_k - \xi^*)$ can be thought of as a kinetic energy and the function term $f(x_k) - f(x^*)$ can be interpreted as a potential energy. For Nesterov's method, one can show that the total energy (or Hamiltonian) decreases at every step although the kinetic energy itself may not decrease in that way.

How to construct a Lur'e-Postnikov Lyapunov function? Apply the dissipation inequality approach with a new supply rate! Recall that Nesterov's method can be written as

$$\begin{aligned}\xi_{k+1} &= A\xi_k + Bu_k \\ v_k &= C\xi_k \\ u_k &= \nabla f(v_k)\end{aligned}\tag{12.1}$$

where $A = \begin{bmatrix} (1+\beta)I & -\beta I \\ I & 0 \end{bmatrix}$, $B = \begin{bmatrix} -\alpha I \\ 0 \end{bmatrix}$, and $C = [(1+\beta)I \quad -\beta I]$. The convergence rate proof of Nesterov's method can be done by applying the dissipation inequality routine presented in past lectures.

1. Replace the nonlinear equation $u_k = \nabla f(v_k)$ in (12.1) by some quadratic inequality in the following form:

$$\begin{aligned}\begin{bmatrix} \xi_k - \xi^* \\ u_k \end{bmatrix}^\top X \begin{bmatrix} \xi_k - \xi^* \\ u_k \end{bmatrix} &\leq -(f(x_{k+1}) - f(x^*)) + \rho^2(f(x_k) - f(x^*)) \\ &= \rho^2(f(x_k) - f(x_{k+1})) + (1 - \rho^2)(f(x^*) - f(x_{k+1}))\end{aligned}$$

The key issue is how to figure out X .

2. Test if there exists $P \geq 0$ such that

$$\begin{bmatrix} A^\top P A - \rho^2 P & A^\top P B \\ B^\top P A & B^\top P B \end{bmatrix} - X \leq 0. \quad (12.2)$$

If so, then the following inequality holds

$$(\xi_{k+1} - \xi^*)^\top P (\xi_{k+1} - \xi^*) - \rho^2 (\xi_k - \xi^*)^\top P (\xi_k - \xi^*) \leq \begin{bmatrix} \xi_k - \xi^* \\ u_k \end{bmatrix}^\top X \begin{bmatrix} \xi_k - \xi^* \\ u_k \end{bmatrix}$$

which is exactly the so-called dissipation inequality $V_{k+1} - \rho^2 V_k \leq S(\xi_k, u_k)$ if we define $V_k = (\xi_k - \xi^*)^\top P (\xi_k - \xi^*)$ and $S(\xi_k, u_k) = \begin{bmatrix} \xi_k - \xi^* \\ u_k \end{bmatrix}^\top X \begin{bmatrix} \xi_k - \xi^* \\ u_k \end{bmatrix}$. Clearly $V_k \geq 0$ due to the fact $P \geq 0$. One can verify that (12.2) holds with $(\rho^2, \alpha, \beta) = (1 - \sqrt{\frac{m}{L}}, \frac{1}{L}, \frac{\sqrt{L} - \sqrt{m}}{\sqrt{L} + \sqrt{m}})$ if P is chosen properly.

3. Now directly apply the supply rate condition to conclude $V_{k+1} + f(x_{k+1}) - f(x^*) \leq \rho^2 (V_k + f(x_k) - f(x^*))$. This rate result can be converted into an ε -optimal iteration complexity result $O(\sqrt{\frac{L}{m}} \log \frac{1}{\varepsilon})$.

How to construct the supply rate condition for Nesterov's method? In the following derivation, write $w_k = u_k = \nabla f(v_k)$. As mentioned previously, if we can construct a symmetric matrix X such that the following supply rate condition holds

$$\begin{aligned} S(\xi_k, w_k) &= \begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix}^\top X \begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix} \\ &\leq -(f(x_{k+1}) - f(x^*)) + \rho^2 (f(x_k) - f(x^*)) \\ &= \rho^2 (f(x_k) - f(x_{k+1})) + (1 - \rho^2) (f(x^*) - f(x_{k+1})), \end{aligned} \quad (12.3)$$

then the dissipation inequality $V(\xi_{k+1}) - \rho^2 V(\xi_k) \leq S(\xi_k, w_k)$ will directly lead to the desired convergence bound $V(\xi_{k+1}) + f(x_{k+1}) - f(x^*) \leq \rho^2 (V(\xi_k) + f(x_k) - f(x^*))$. The key issue is how to figure out X . If we can find X_1 and X_2 such that

$$\begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix}^\top X_1 \begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix} \leq f(x_k) - f(x_{k+1}) \quad (12.4)$$

$$\begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix}^\top X_2 \begin{bmatrix} \xi_k - \xi^* \\ w_k \end{bmatrix} \leq f(x^*) - f(x_{k+1}), \quad (12.5)$$

then we can set $X = \rho^2 X_1 + (1 - \rho^2) X_2$ to obtain the condition (12.3). Now let's look at how to obtain X_1 and X_2 .

For illustrative purposes, we focus on the construction of X_1 . The construction of X_2 will be similar. The condition (12.4) involves $f(x_{k+1})$ and $f(x_k)$. Hence it is reasonable to

think that its construction requires some inequalities involving the function value f . Recall that L -smoothness and m -strong convexity give the following two inequalities:

$$f(x) \leq f(y) + \nabla f(y)^\top (x - y) + \frac{L}{2} \|x - y\|^2 \quad (12.6)$$

$$f(x) \geq f(y) + \nabla f(y)^\top (x - y) + \frac{m}{2} \|x - y\|^2 \quad (12.7)$$

How can we choose (x, y) in the above inequalities to obtain (12.4). One idea is to set $(x, y) \rightarrow (x_{k+1}, x_k)$ in (12.6). However, $\nabla f(y)$ becomes $\nabla f(x_k)$ and this is not w_k ! The only term involving the gradient information on the left side of (12.4) is w_k which is the gradient evaluated on v_k ! Therefore, when applying (12.6) and (12.7) to construct (12.4), one has to set y to be v_k ! By doing this, we can show

$$\begin{aligned} f(x_k) - f(x_{k+1}) &= f(x_k) - f(v_k) + f(v_k) - f(x_{k+1}) \\ &\geq \nabla f(v_k)^\top (x_k - v_k) + \frac{m}{2} \|x_k - v_k\|^2 \\ &\quad + \nabla f(v_k)^\top (v_k - x_{k+1}) - \frac{L}{2} \|v_k - x_{k+1}\|^2 \\ &= \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix}^\top \left(\frac{1}{2} \begin{bmatrix} \beta^2 m & -\beta^2 m & -\beta \\ -\beta^2 m & \beta^2 m & \beta \\ -\beta & \beta & \alpha(2 - L\alpha) \end{bmatrix} \otimes I_p \right) \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix} \end{aligned}$$

The last step in the above derivation requires substituting $x_{k+1} = (1 + \beta)x_k - \beta x_{k-1} - \alpha \nabla f(v_k)$ and $v_k = C\xi_k$ into the second-to-last line $\nabla f(v_k)^\top (x_k - v_k) + \frac{m}{2} \|x_k - v_k\|^2 + \nabla f(v_k)^\top (v_k - x_{k+1}) - \frac{L}{2} \|v_k - x_{k+1}\|^2$ and rewriting the resultant quadratic function. This gives us the matrix X_1 . We can see that the key trick is just subtracting and adding $f(v_k)$.

Similarly, X_2 can be derived as

$$\begin{aligned} f(x^*) - f(x_{k+1}) &= f(x^*) - f(v_k) + f(v_k) - f(x_{k+1}) \\ &\geq \nabla f(v_k)^\top (x^* - v_k) + \frac{m}{2} \|x^* - v_k\|^2 \\ &\quad + \nabla f(v_k)^\top (v_k - x_{k+1}) - \frac{L}{2} \|v_k - x_{k+1}\|^2 \\ &= \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix}^\top X_2 \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \\ \nabla f(v_k) \end{bmatrix} \end{aligned}$$

The details of X_2 are left as an exercise.